

Tukun's Classes

$(G, \cdot)$, ① $|H| < \infty$.

② H is closed under $\cdot$.

Show that $H \leq G$.

$\uparrow$
Subgroup on equal.

poset [partially ordered set]

(X, P) 

P is called partial order relation
iff $\rightarrow$

① Reflexive

② Transitive

③ ~~Symmetric~~ Anti-symmetric

$a P b \not\Rightarrow b P a$


$a P b \Leftrightarrow b P a$
 $\Rightarrow a = b$

$$(\mathbb{R}, \leq)$$

↓

$$a \leq b \Rightarrow b \leq a$$

$$1 \leq 2 \Rightarrow 2 \not\leq 1$$

$$a \leq b \Leftrightarrow b \leq a$$

$$\underline{a = b}$$

$$(\mathcal{P}(X), \subseteq)$$

$$A \subseteq B \not\Leftrightarrow B \subseteq A$$

$$A \subseteq B \Leftrightarrow B \subseteq A$$

$$\Rightarrow A = B.$$

$G \rightarrow$

$\text{Subgr}(G) = \text{set of all subgroups of } G.$

$$(\underline{\text{Subgr}(G)}, \leq) \rightarrow \underline{\text{poset}}$$



partial order relation

$A \in \text{Subgr}(G)$

$B \in \text{Subgr}(G)$

A subgr. of G , B subgr. of G .

$\Rightarrow A \leq B \Rightarrow A$ subgr of B .

$H \in \text{Subgr}(G)$, $H \leq G$.

$A \leq G$.

$H \leq G$

pf: Test-①

$a, b \in H \subseteq G$

$\Rightarrow \boxed{ab^{-1}} \in H$; $b^{-1} \in G$.

$b^{-1} \stackrel{?}{\in} H$

$a = e$

$e \cdot b^{-1} \in H$

$\Rightarrow b^{-1} \in H$

Subgroup means: $(H \leq G)$ $[G, \cdot]$

✓① H is subset of G ($H \subseteq G$)

✓② $(H, \cdot)$ is a group under $\cdot$.

Ex: $(\mathbb{Q}, \cdot)$ group

✓ ① $\mathbb{Z} \subseteq \mathbb{Q}$

✓ ② $(\mathbb{Z}, \cdot)$ is ~~group~~ or not.

$(\mathbb{Z}, +)$ is group $\times$

$$(\mathbb{Q}, +) \rightarrow (\mathbb{Z}, +) \leq (\mathbb{Q}, +)$$

→ binary operation

$$\cdot : \underline{G \times G} \rightarrow G_2$$

closure.

function

$$(a, b) \mapsto a \cdot b$$

$$((G \times G) \times G, \cdot)$$

$$(a, b) \rho a \cdot b$$

$$((a, b), a \cdot b)$$

$$\begin{array}{c} a \rho b \quad b \neq c \\ a \rho c \\ \hline b = c \end{array}$$

$$(a, b) \mapsto a \cdot b$$

$$x = a \cdot b$$

Binary operation: $[A, B, \overset{\text{any}}{C} \text{ are sets}]$

$$f : A \times B \rightarrow C$$

$$(a, b) \mapsto c$$

$$f(a, b) = c$$

$$\Rightarrow a \cdot b = c$$

Test ① $H \subseteq G$

① If $ab \in H$ whenever $a, b \in H$

② If $a^{-1} \in H$, whenever $a \in H$

Finite Subgr. Test

pf: $a \in H \neq \varnothing$

$$\{\underbrace{a^1, a^2, a^3, a^4, a^5, \dots}_{a \in H}\} \subseteq H$$

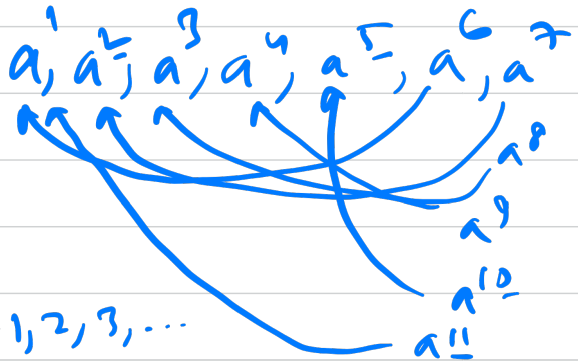
$$a^2 = a \cdot a \in H$$

$$a^3 = a^2 \cdot a \in H$$

$$a^i = a^j, \quad i, j = 1, 2, 3, \dots$$

$$\Rightarrow a^i (a^j)^{-1} = a^j (a^i)^{-1}$$

$$[a^i \in H \subseteq G, (a^j)^{-1} \in G]$$



$$\Rightarrow a^i (a^{-1})^j = e \quad [i > j \text{ (wlog)}]$$

$$\Rightarrow a^{i-j} = e \quad \text{---} \textcircled{*}$$

$$\underline{b \cdot (a^j)^{-1} = e}$$

$$\Rightarrow b = a^j = \underbrace{a \cdot a \cdot \dots \cdot a}_j$$

$$\Rightarrow b a^{-1} = a \cdot \dots \cdot \underbrace{a}_{j} a^{-1}$$

$$\Rightarrow b \underbrace{(a^{-1})(a^{-1})(a^{-1}) \dots (a^{-1})}_{j} = e$$

$$\Rightarrow b \underbrace{(a^{-1})^j}_j = e$$

$$\Rightarrow \underline{(a^j)^{-1}} = \underline{(a^{-1})^j}$$

$$a^i (a^{-1})^j$$

$$= \underbrace{a \cdot \dots \cdot a}_i \cdot \underbrace{a^{-1} \cdot \dots \cdot a^{-1}}_j$$

$$= \underbrace{a \cdot \dots \cdot a}_{i-j} = a^{i-j}$$

$$\text{From } \textcircled{*} \quad a^{i-j} = e$$

$$\Rightarrow a \cdot a^{\overbrace{i-j-1}} = e$$

$$i > j \quad \checkmark$$

$$\Rightarrow i \geq j+1$$

$$\Rightarrow \overbrace{i-j-1} \geq 0$$

$$2 > 1$$

$$2 > 1+1$$

$$\Rightarrow a a^k = e$$

$$\begin{aligned} [k &= i-j-1 \\ \Rightarrow k &\geq 0] \end{aligned}$$

case (i) $k \neq 0, k > 0$.

$$a a^k = e \Rightarrow \underbrace{a^k}_{\neq} = a^{-1} \in G.$$

$$H \Rightarrow \underline{a^{-1} \in H}$$

case (ii) $k = 0$,

$$a = e \Rightarrow a^{-1} = e^{-1} = e.$$

$$\therefore a \in H$$

$$\Rightarrow \underline{e} \in H \Rightarrow a^{-1} = e \Rightarrow a^{-1} \in H.$$

$\langle a \rangle$ is a Subgroup, $a \in G$.

pf. $\langle a \rangle = \{ a^n \mid n \in \mathbb{Z} \}$

angle bracket = $\{ \dots, a^{-2}, a^{-1}, a^0, a^1, a^2, \dots \}$

Test (II) : $x, y \in \langle a \rangle$ Mystery

$$\Rightarrow x = a^m, y = a^n, n, m \in \mathbb{Z}$$

$$\Rightarrow xy = a^m a^n = a^{m+n} \in \langle a \rangle$$

Isomorphism:

$$\begin{aligned} & (\langle a \rangle, \cdot) \rightarrow (\mathbb{Z}, +) \\ & (\langle a \rangle, \cdot) \rightarrow (\mathbb{Z}_m, +) \end{aligned}$$

$$\because n+m \in \mathbb{Z}$$

Involve: By definition.

$\langle a \rangle$ = cyclic subgroup of G generated by a .

$$a^m \cdot a^n = a^{\overline{m+n}} = a^{\overline{n+m}} = a^n a^m$$

To show $\langle a \rangle$ is Abelian (commutative)
we need $(\mathbb{Z}, +)$ to be Abelian.